

(notes) 3.7 Inverse Functions

Original function: $f(x)$

Inverse function: $f^{-1}(x)$

Key Takeaways

1. Algebra: when you apply the inverse operation, you get back the identity, x
2. Graphically: when you plot a function $f(x)$ and its inverse function $f^{-1}(x)$, the two functions are graphically symmetric over the identity function, $y = x$!
3. Functions need to be one-to-one to have an inverse. If not, we need to restrict the domain of $f(x)$.

Algebraic check!

$f(x)$	$f^{-1}(x)$	$f^{-1}(f(x)) = x?$ (check if we get the identity, x)
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$$x + 2$$

$$2x$$

$$x^2$$

$$2^x$$

$$\log_2(x)$$

$$f^{-1}(f(x)) = \log_2(f(x)) = \log_2(2^x) = x \checkmark$$

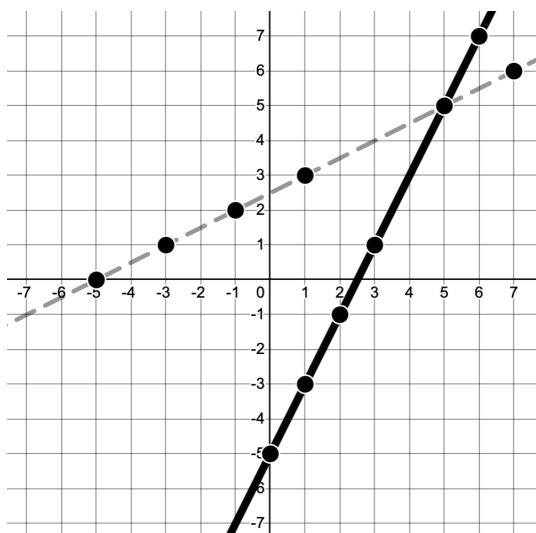
$$2x - 5$$

$$\frac{x+5}{2}$$

$$f^{-1}(f(x)) = \frac{f(x) + 5}{2}$$

(2x - 5) + 5 = 2x

Graph check!



Solid: $f(x) = 2x - 5$

Dashed: $f^{-1}(x) = \frac{x+5}{2}$

original function

$(x, f(x))$

$(0, -5)$

$(1, -3)$

$(2, -1)$

$(3, 1)$

$(5, 5)$

$(6, 7)$

inverse function

$(x, f^{-1}(x))$

$(-5, 0)$

$(-3, 1)$

$(-1, 2)$

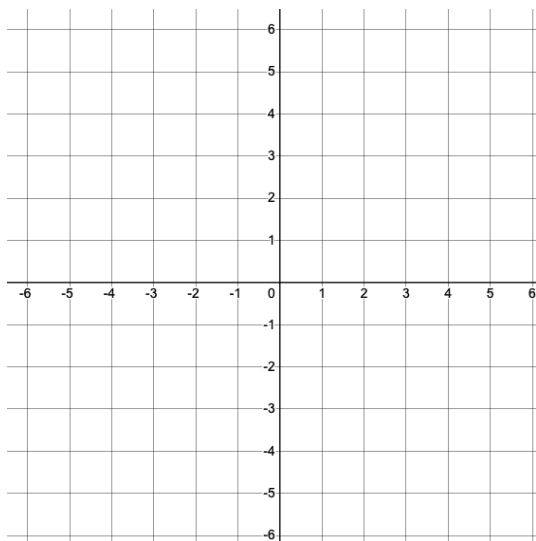
$(1, 3)$

$(5, 5)$

$(7, 6)$

First, graph $f(x)$ and $f^{-1}(x)$.

Then, fill in the table with the coordinate values for each of the functions.



Graph $f(x) = 2x$

Graph $f^{-1}(x) = \frac{1}{2}x$

original function

$(x, f(x))$

$(-2, \quad)$

$(-1, \quad)$

$(0, \quad)$

$(1, \quad)$

$(2, \quad)$

inverse function

$(x, f^{-1}(x))$

$(\quad, -2)$

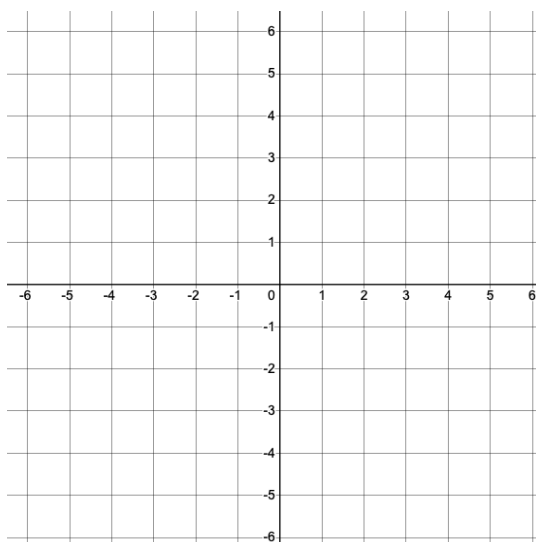
$(\quad, -1)$

$(\quad, 0)$

$(\quad, 1)$

$(\quad, 2)$

Graph check!



What is odd about this one???

Graph $f(x) = x^2$

Graph $f^{-1}(x) = \sqrt{x}$.

original function

$(x, f(x))$

inverse function

$(x, f^{-1}(x))$

$(-2, \quad)$

$(\quad, -2)$

$(-1, \quad)$

$(\quad, -1)$

$(0, \quad)$

$(\quad, 0)$

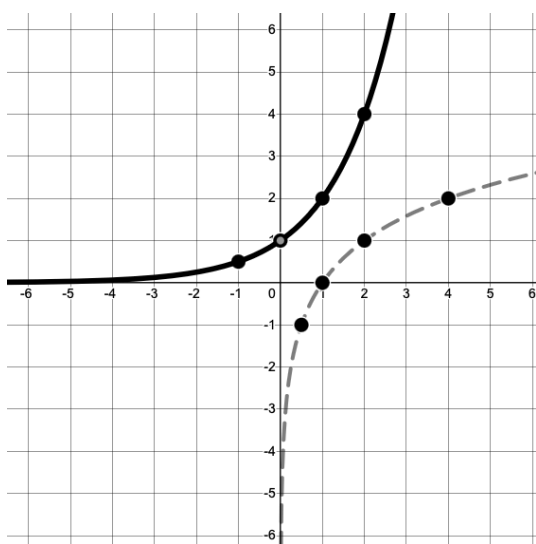
$(1, \quad)$

$(\quad, 1)$

$(2, \quad)$

$(\quad, 2)$

Use the graph to fill in the tables with indicated points



Solid: $f(x) = 2^x$

Dashed: $f^{-1}(x) = \log_2(x)$.

original function

$(x, f(x))$

inverse function

$(x, f^{-1}(x))$

$(\quad, \quad)$

$(\quad, \quad)$

$(\quad, \quad)$

$(\quad, \quad)$

$(\quad, \quad)$

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$(\quad, \quad)$

$(\quad, \quad)$

Discuss: how are the graphs of $f(x)$ and $f^{-1}(x)$ related? How are the coordinate points related?

Practice with tables

Question 2

Use the table below to fill in the missing values.

x	$f(x)$
0	7
1	1
2	9
3	4
4	8
5	6
6	3
7	0
8	5
9	2

answers: a) 4, b) 5, c) 3, d) 6

a. $f(3) =$

b. If $f(x) = 3$, then $x =$

c. $f^{-1}(4) =$

d. If $f^{-1}(x) = 5$, then $x =$

Solving for an inverse

To solve for the inverse, swap $y \leftrightarrow x$. Then solve for y .

Example. Solve for $f^{-1}(x)$

$$f(x) = 7x + 9$$

$$y = 7x + 9$$

$$\text{inverse: } y \leftrightarrow x$$

$$x = 7y + 9 \quad \rightarrow \text{now solve for } y$$

$$x - 9 = 7y + 9 - 9 \quad \rightarrow \text{subtract 9}$$

$$x - 9 = 7y$$

$$\frac{x - 9}{7} = \frac{7y}{7} \quad \rightarrow \text{divide both sides by 7}$$

$$\frac{x - 9}{7} = y \quad \implies f^{-1}(x) = \frac{x - 9}{7}$$

Practice, solve for $f^{-1}(x)$

1. $f(x) = 14 - x$

2. $f(x) = 2x + 5$

3. $f(x) = (x - 6)^2$

(.....is this function one-to-one and if not, how can we **restrict the domain** to just the non-decreasing section?)

4. $f(x) = \frac{1}{x+10}$

5. $f(x) = 12 + \sqrt[3]{x}$